

Daniel Haas

Manual: P.H.A.A.S.T

Universidade Federal do Rio de Janeiro
2025

1 Usage

PHAAST has two main non-optional and positional arguments, `STOICHIOMETRY`, which defines the atoms that will compose the searched minima, and `OUTPUT_PREFIX`, a starting name given to the resulting minima that will be stored in xyz format. The command can be summarized in:

```
1 python -m phaast [OPTIONS] STOICHIOMETRY OUTPUT_PREFIX
```

1.1 Options

All other option arguments can be found on the help section, which is shown with the use of the option `-h`:

```
1 -h, --help                show this help message and exit
2
3 --return_number RETURN_NUMBER
4   Number of least energy molecules to return at the end of the algorithm (
   default = 50)
5
6 -pop, --population_size POPULATION_SIZE
7   Number of molecules in the population (default = 5000)
8
9 -c, --charge CHARGE       Structure's charge (default = 0)
10
11 -t, --threads THREADS
12   Number of threads to be used by the algorithm (default = number of
   threads on cpu)
13
14 -xtbt, --xtb-threads XTB_THREADS
15   Number of threads to be used by each -t xtb instance (default = 4)
16
17 --xtb_path XTB_PATH
18   Path to be used for the xtb binary (default = xtb)
19
20 -et, --energy_threshold ENERGY_THRESHOLD
21   Maximum energy difference [in hartree] so molecules are considered alike
   (default = 0.001)
22
23 -gt, --geometry_threshold GEOMETRY_THRESHOLD
24   Maximum geometry difference so molecules are considered the same so one
   of them is discarded [must be a value between 0 and 1] (default = 0.87)
25
26 --mut_disp_w MUT_DISP_W
27   Mutation displacement weight (default = 3)
28
29 --mut_perm_w MUT_PERM_W
30   Mutation permutation weight (default = 2)
31
32 --mut_twist_w MUT_TWIST_W
33   Mutation twist weight (default = 1)
34
35 --crov_w CROV_W
36   Crossing-over weight (default = 1)
37
38 --migr_w MIGR_W
39   Migrator weight (default = 1)
40
41 --mut_disp_min MUT_DISP_MIN
42   Minimum distance (A) to be used in mutation of atomic displacement (
   default = 1.3)
43
44 --mut_disp_max MUT_DISP_MAX
```

```

45 Maximum distance (A) to be used in mutation of atomic displacement (
    default = 2.3)
46
47 --mut_disp_num MUT_DISP_NUM
48 Number of atoms to move in mutations of atomic displacement (default =
    1)
49
50 --mut_perm_num MUT_PERM_NUM
51 Number of permutations in mutations of that type (default = Half the
    number of atoms)
52
53 --mut_twist_min_angle MUT_TWIST_MIN_ANGLE
54 Minimum angle to be used in twist mutations (default = 90)
55
56 --mut_twist_max_angle MUT_TWIST_MAX_ANGLE
57 Maximum angle to be used in twist mutations (default = 270)
58
59 -loops, --end_loop_number END_LOOP_NUMBER
60 Number of loops the algorithm will run with the least energy molecule
    until a new one is found, terminating it (default = 11)
61
62 --comparison_algorithm COMPARISON_ALGORITHM
63 Algorithm to be used for structures comparison, the recommends usage is:
64 - "grigoryan_springborg" for clusters;
65 - "bonding_length" for organic and general shaped structures (check --
    comparison_bonding_tolerance too);
66 (default = grigoryan_springborg)
67
68 --comparison_bonding_tolerance COMPARISON_BONDING_TOLERANCE
69 Distance tolerable so two atoms can be considered bonded within the
    molecule comparison algorithm and bonding tests
70
71 --remove_unbonded REMOVE_UNBONDED
72 Choose to not remove unbonded structures from genetic algorithm
    population
73 - "always": Remove unbonded in every loop;
74 - "final": Remove only from final population to not contaminate
    results;
75 - "never": Do not remove;
76 (default = always)
77
78 -v, --verbose

```

1.1.1 Return number

Changing the `return_number` will return more or less structures at the end of the algorithm, there's no additional overhead to that, since the structures have already been optimized up to that point. It is useful when the calculator used to optimize the structures is not so accurate on the energy values and a better optimization (on a more refined functional/basisset) is needed at the end of the algorithm, hence justifying a smaller return number.

1.1.2 Population size

Despite how it sounds, this argument doesn't define a cap or a direct control on the population's size, it actually defines how many structures will be generated in each generation, respecting the **Weights** defined at the optional arguments.

1.1.3 Threads

Number of processes (or threads if you are using a GIL-disabled python instance) used for paralelizing the expensive tasks on the genetic algorithm.

1.1.4 Weights

All the `--*_w` arguments define the proportions of how each operation will be applied. For instance, `--mut_disp_w`, `--mut_perm_w` and `--mut_twist_w` define all mutation weights, if they're the same value, all mutations will occur in the same proportion.